Why Algorithms Need the Machine

Locality, Speed, and the 16-Structure Sprint

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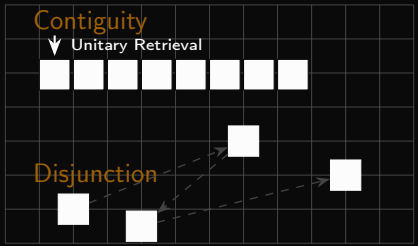
The Spatial Premise



- Modern CPUs operate on a **spatial premise**.
- Contiguous layouts leverage hardware prefetching.
- Pointer-chasing induces serial stalls.
- Memory latency is the **primary bottleneck** of the 21st-century.

The Scale of the Project

16
Data Structural Families

6,641
Datapoints

730B
Memory Elements Analyzed

The Reality: Locality is a Multiplier on Big-O

Big-O defines the *curve*. Hardware locality defines the *crossover point*.

> *"Memory Access is like a Coffee Shop vs. a Warehouse"*

L1 Cache (Coffee Shop)
Right in front of you. Fast. Automatic.

RAM (The Warehouse)
Slow. Requires a trip. CPU stalls while waiting.

For operations with identical complexity, spatial locality dictates the winner.

Meet the Contenders: Beyond the BST

How do we organize data when a simple list isn't enough?

SkipList

The Express Elevator

A list with "shortcut" lanes to skip nodes. Fast as a Tree, but simpler.

vEB Tree

Digital Fractals

Uses recursion to divide data into clusters. Theoretically the fastest.

B-Tree

The Librarian

Stores hundreds of items per node. Keeps the CPU happy by minimizing trips.

Measured all 16 contenders across 5 orders of magnitude.

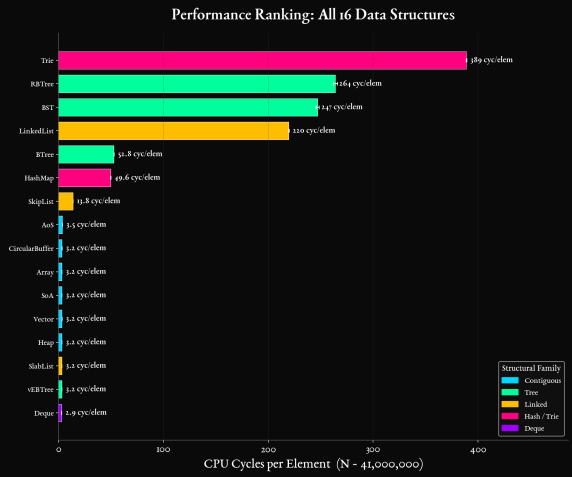
Methodology: The Precision Toolkit

```
// CPU Pinning
taskset -c 1 ./benchmark

// Memory Barrier
__asm__ __volatile__("" ::: "
    memory");

// Hardware PMU Access
syscall(SYS_perf_event_open,
    ...);

// Optimization Shield
volatile long long sink = 0;
```

The Scientific Rigor:

- **taskset:** Core isolation.
- **asm/volatile:** Prevents compiler 'magic'.
- **perf:** True cycle counts.

Who Wins the Sprint? (N = 41 Million)

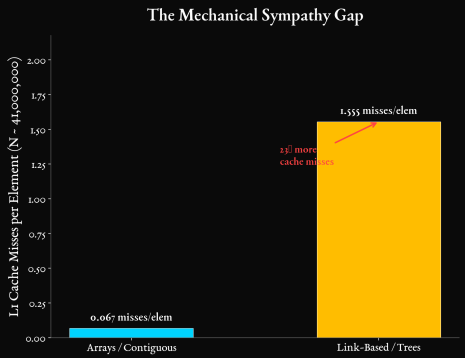


The Traversal Bound

Arrays win by 40x *during full scans.*

But remember: inserting into an array costs $O(N)$ data movement.

The Secret Multiplier: Locality



> *"Locality is like Shopping in one aisle vs. running across the store."*

Pointer Tax: Links or Pointers force the CPU to find a new address for every single element.

Practical Lessons

Algorithms are Math, but Performance is Physics.

I. **Contiguity is King (for Retrieval):** Locality maximizes throughput.

II. **The Write Penalty:** Beware $O(N)$ data movement during mutations.

III. **The B-Tree Compromise:** Balance Big-O efficiency with cache contiguity.

Any Questions?

Thank You!

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